

A REPORT
ON
**CUSTOMER CHURN PREDICTION USING MACHINE
LEARNING**

Submitted by

Ms. Shaziya Banu
IS-2026-9761

INTERNSPARK

ARTIFICIAL INTELLIGENCE INTERNSHIP

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1. Introduction

Customer churn refers to customers discontinuing a company's services. Predicting customer churn is important because retaining existing customers is often more cost-effective than acquiring new customers.

The objective of this project is to build and evaluate machine learning models capable of predicting whether a customer is likely to churn based on demographic information, account details, and service usage patterns.

Two machine learning algorithms were implemented and compared:

1. Logistic Regression
2. Random Forest Classifier

The model performance was evaluated using various classification metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

2. Dataset Description

The IBM Telco Customer Churn dataset was used for this project.

Dataset Characteristics:

- Total Records: 7043
- Total Features: 21
- Target Variable: Churn
- Problem Type: Binary Classification

Important Features:

- Gender
- SeniorCitizen
- Partner
- Dependents
- Tenure
- PhoneService
- InternetService
- Contract
- PaymentMethod
- MonthlyCharges
- TotalCharges

5 rows x 21 columns

Downloaded from <http://ajph.org/> on November 10, 2015

Target Variable:

- Churn = Yes (Customer left the service)
- Churn = No (Customer retained)

3. Data Preprocessing

Several preprocessing steps were performed before model training.

3.1 Handling Missing Values

The TotalCharges column contained missing values.

Steps performed:

- Converted TotalCharges from string to numeric format.
- Missing values were replaced using the median value of the column.

3.2 Removing Irrelevant Features

The customerID column was removed because it does not contribute to churn prediction.

3.3 Encoding Categorical Variables

Machine learning algorithms require numerical inputs.

Therefore, categorical variables such as:

- Gender
- Partner
- Contract
- PaymentMethod
- Churn

were converted into numerical values using Label Encoding.

4. Exploratory Data Analysis (EDA)

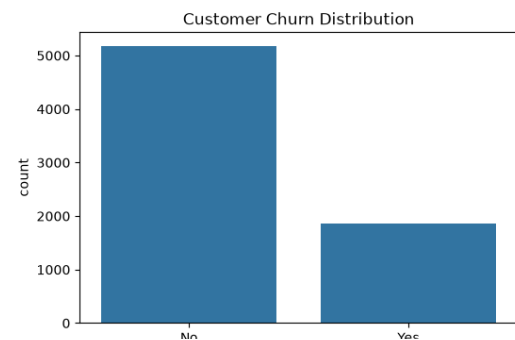
Exploratory Data Analysis was conducted to understand the dataset and identify important patterns.

4.1 Customer Churn Distribution

A count plot was created to visualize the distribution of churned and retained customers.

Observation:

Most customers were retained, while a smaller percentage had churned.

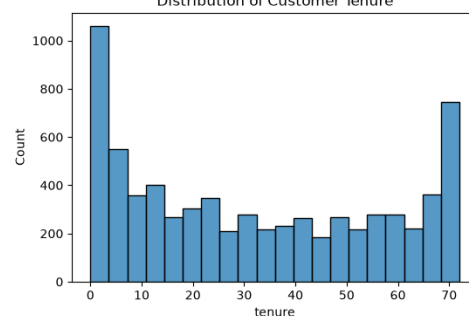


4.2 Customer Tenure Distribution

A histogram was used to analyze customer tenure.

Observation:

Many customers had shorter tenures, indicating a large number of recently acquired customers.

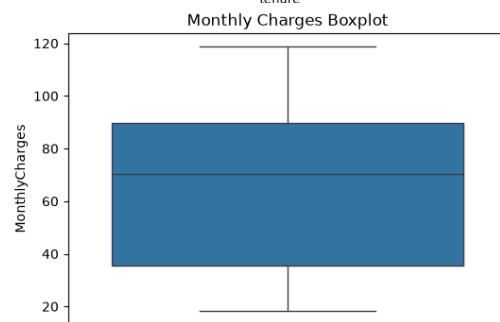


4.3 Monthly Charges Analysis

A boxplot was created for MonthlyCharges.

Observation:

The dataset contains a wide range of monthly charges with a few outliers.

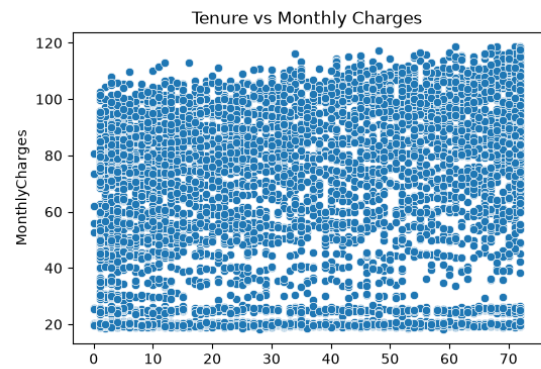


4.4 Tenure vs Monthly Charges

A scatter plot was generated to analyze the relationship between tenure and monthly charges.

Observation:

Customers with different tenure values exhibited varying monthly charges, indicating diverse service plans.

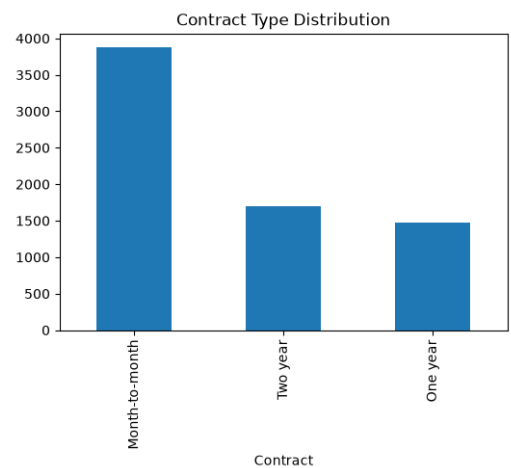


4.5 Contract Type Distribution

A bar chart was created to examine contract types.

Observation:

Month-to-month contracts were the most common contract type.



5. Feature Selection

The dataset was divided into:

Independent Variables (X): All features except Churn.

Dependent Variable (y): Churn.

The final dataset contained 19 input features.

6. Train-Test Split

The dataset was split into training and testing sets.

Configuration:

- Training Data: 80%
- Testing Data: 20%
- Random State: 42

This ensures that model evaluation is performed on unseen data.

7. Model Building

7.1 Logistic Regression

Logistic Regression is a statistical classification algorithm commonly used for binary classification problems.

Advantages:

- Simple and interpretable

- Fast training
- Effective for binary classification

7.2 Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees.

Advantages:

- Handles complex relationships
- Reduces overfitting
- High predictive performance

8. Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

```
Logistic Regression
-----
Accuracy : 0.8168914123491838
Precision: 0.6802507836990596
Recall    : 0.5817694369973191
F1 Score  : 0.6271676300578035
ROC-AUC   : 0.8615253035494322
```

```
Random Forest
-----
Accuracy : 0.7955997161107168
Precision: 0.6591760299625468
Recall    : 0.4718498659517426
F1 Score  : 0.55
ROC-AUC   : 0.8369592783131656
```

9. Model Comparison

Comparison of both models indicates that Logistic Regression outperformed Random Forest across all evaluation metrics.

Key Findings:

- Higher Accuracy
- Better Precision
- Better Recall
- Better F1 Score
- Higher ROC-AUC Score

Therefore, **Logistic Regression** was selected as the final model.

10. Cross Validation

To verify model stability and generalization, 5-Fold Cross Validation was performed.

Cross Validation Scores:

- 0.8062

- 0.8112
- 0.7864
- 0.8075
- 0.8040

Cross Validation Scores:
[0.80624556 0.81121363 0.78637331 0.80752841 0.80397727]

Average CV Accuracy:
0.8030676374282212

Average Cross Validation Accuracy:

80.31%

Observation:

The cross-validation results show consistent performance across multiple folds, indicating that the model generalizes well and is not overfitting.

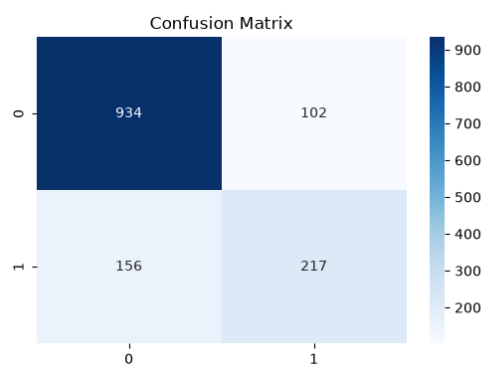
11. Confusion Matrix

A confusion matrix was generated to evaluate classification performance.

The confusion matrix provides:

- True Positives
- True Negatives
- False Positives
- False Negatives

This helps in understanding prediction errors and model effectiveness.

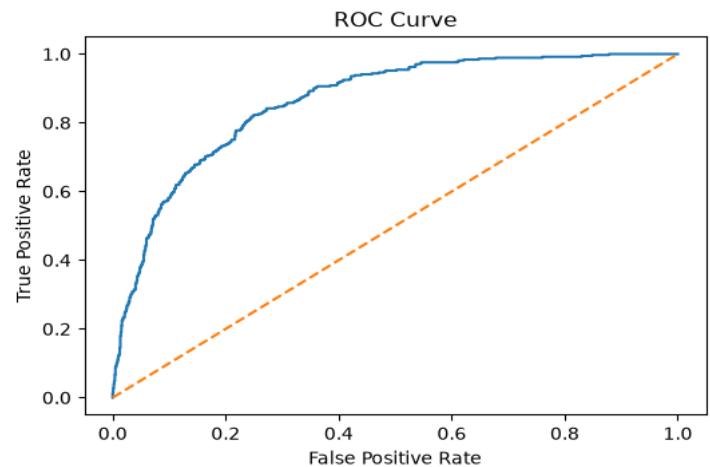


12. ROC Curve Analysis

The ROC Curve was plotted to analyze the trade-off between:

- True Positive Rate
- False Positive Rate

The Logistic Regression model achieved a ROC-AUC score of 0.8615, indicating strong classification capability.



	precision	recall	f1-score	support
0	0.86	0.90	0.88	1036
1	0.68	0.58	0.63	373
accuracy			0.82	1409
macro avg	0.77	0.74	0.75	1409
weighted avg	0.81	0.82	0.81	1409

13. Conclusion

In this project, a machine learning solution was developed to predict customer churn using the IBM Telco Customer Churn dataset.

The dataset was cleaned, preprocessed, analyzed, and used to train two classification models.

Among the evaluated models, Logistic Regression achieved the best overall performance with:

- Accuracy: 81.69%
- F1 Score: 62.72%
- ROC-AUC: 86.15%

Cross-validation further confirmed the reliability and robustness of the model.

The results demonstrate that Logistic Regression provides a strong balance between simplicity and predictive performance, making it the preferred model for this customer churn prediction task.

The final model can be deployed as a prediction API and further analyzed using explainable AI techniques such as SHAP and LIME.

Technologies Used

- Python 3.11
- NumPy
- Pandas
- Matplotlib

- Seaborn
- Scikit-Learn
- Jupyter Notebook

References

1. IBM Telco Customer Churn Dataset
2. Scikit-Learn Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Matplotlib Documentation